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INTEGRAL ELEMENTS OF OKUBO ALGEBRA AND THE E8-LATTICE

DANIELE CORRADETTI

Abstract. In this work we study the interplay between the Coxeter-Dickson E_8-order, the para-octonions, and the real Okubo algebra. We first explain why a usual rotation presentation of the order-three automorphism, although algebraically correct, does not preserve the classical integral orders. We then introduce an integral order-three automorphism and prove that its Petersson isotope is the compact real Okubo algebra. The four octonionic orders with lattices C_8, A_2^4, D_4^2, and E_8 are preserved by the new realisation of the Okubo product. In particular, the same Coxeter-Dickson E_8 lattice supports octonionic, para-octonionic, and Okubo integral structures, although the resulting multiplicative systems are not isomorphic.

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introduction and motivations

Integral numbers in division algebras are a classical and fascinating topic at the *interplay* of algebra, arithmetic, and exceptional geometry. Already in the XIX century, Gauss [Gau] and Hurwitz [Hu19] extended the notion of integers to the complex numbers and the quaternions, identifying the Gaussian integers and the Hurwitz quaternions as natural integral structures. The culmination of this classical line of research is the *Coxeter-Dickson octonions* $\mathbb{O}_{E_8}$ [Co46, Di23], the maximal order in the algebra of the octonions $\mathbb{O}$, which is isometric to the exceptional lattice E_8. This octonionic background is also close to modern exceptional geometry and to the systematic study of octonionic Rosenfeld spaces [MCCAI23].

The landscape of eight-dimensional real division algebras, however, extends well beyond the octonions. As a consequence of the Generalised Hurwitz Theorem [El18], among the eight-dimensional composition algebras over $\mathbb{R}$ there are three non-isomorphic division cases: the *octonions* $\mathbb{O}$ (unital), the *para-octonions* $p\mathbb{O}$ (para-unital), and the *real Okubo algebra* $\mathcal{O}$ (non-unital). While octonions have a unit element and para-octonions have a para-unit, the Okubo algebra merely contains *idempotent* elements, and its automorphism group is SU(3) rather than G_2. A synoptic comparison of these algebras is given in Table 3. This non-unital direction has recently appeared in the study of Okubo projective geometry, Okubo Spin groups, minimal

Cayley-plane realisations, collineation groups of octonionic planes, and possible physical uses of the Okubo algebra [CZ22, CMZ24, CMZ25, MCZ25].

The question of integral elements for $p\mathbb{O}$ and $\mathcal{O}$ has, to our knowledge, not been systematically separated from the classical unital case. The definition due to Dickson [Di23] and refined by Johnson [Jo13, Jo17, Jo18] requires the presence of a unit element, and therefore does not directly apply to non-unital algebras such as $p\mathbb{O}$ and $\mathcal{O}$. In this work we isolate the part of Johnson's definition that survives for isotopes: closure under the relevant product, integrality of norm and trace, and the relation with a discrete lattice or module. For complementary viewpoints on eight-dimensional composition algebras, the Cayley plane, and their recovery from geometric algebra, see also [Cor26, Cor24].

The para-octonionic case gives a clean result. Since the Coxeter-Dickson order $\mathbb{O}_{E_8}$ is closed under octonionic conjugation and multiplication, it is also closed under $x \bullet y = \overline{x} \cdot \overline{y}$; hence the same E_8 lattice supports a $\mathbb{Z}$-integral para-octonionic structure. This is a perfect analogy with the octonionic case.

The Okubo case is subtler. A common Petersson presentation uses a rotation by $2\pi/3$ and consequently displays coefficients in $\mathbb{Q}(\sqrt{3})$. That automorphism produces the real Okubo algebra, but it does not preserve the standard octonionic orders. The obstruction is therefore arithmetic rather than algebraic. We replace this rotation by an integral signed permutation τ of the Fano basis. Its fixed subalgebra is quaternionic, so the classification of order-three Petersson algebras shows that the resulting isotope is again the compact real Okubo algebra. At the same time, τ preserves all four crystallographic octonionic orders considered here.

The main consequence is simple: the Coxeter-Dickson order itself is closed under the new Okubo product. Thus the octonionic and Okubo constructions have exactly the same additive E_8 lattice, the same norm, and the same 240 vectors of norm one. Their products are nevertheless different: the octonionic roots form a Moufang loop, whereas the Okubo roots form a finite quasigroup with distinguished idempotents.

The work is organised as follows. In Sec. 1 we review the classical notion of integral elements over division algebras following Dickson and Johnson. In Sec. 2 we review the three algebras $\mathbb{O}$, $p\mathbb{O}$, and $\mathcal{O}$ and their interplays. In Sec. 3 we exhibit the failure of the old rotation, present the integral automorphism, and prove preservation of the four integral systems. In Sec. 4 we compare the octonionic and Okubo structures on the common E_8 lattice. Finally, in Sec. 5 we discuss our findings and outline future directions.

1. WHAT IS AN INTEGER

Modern research on integral numbers started in the XIX century with Gauss, Kronecker and Frege, culminating in the works of Dedekind, Peano [De88, Pe89] on the axiomatization of natural and integer numbers. On the other hand, attempts to generalise the notion of integers in the realm of division algebras such as $\mathbb{C}$ complex numbers and $\mathbb{H}$ quaternions were done by Gauss[Gau], Hamilton and Hurwitz [Hu19]. From that chain of research the definition that settled in the scientific community was that of Leonard Dickson in 1923 [Di23, p. 141-142] which we briefly review here. Given a composition algebra $\mathbb{A}$, i.e. an algebra with a norm n such that $n(x \cdot y) = n(x)\, n(y)$, endowed with an involution $x \longrightarrow \overline{x}$ called *conjugation*, then a set I of *integral elements* of $\mathbb{A}$ is a set that is closed under addition and multiplication that contains the element 1 and such that for every element $x \in I$ we have

$$tr(x) = x + \overline{x} \in \mathbb{Z}, \tag{1.1}$$

$$n(x) = x\overline{x} \in \mathbb{Z}. \tag{1.2}$$

Name	Alg.	Dim.	Symbol	Unit El.	Lattice
Integers	$\mathbb{R}$	1	$\mathbb{Z}$	2	A_1
Eisenstein	$\mathbb{C}$	2	$\mathbb{C}_{A_2}$	3	A_2
Gaussian	$\mathbb{C}$	2	$\mathbb{C}_{C_2}$	4	C_2
Hamilton	$\mathbb{H}$	4	$\mathbb{H}_{2C_2}$	8	$C_2 \oplus C_2$
Hybrid	$\mathbb{H}$	4	$\mathbb{H}_{2A_2}$	12	$A_2 \oplus A_2$
Hurwitz	$\mathbb{H}$	4	$\mathbb{H}_{D_4}$	24	D_4
Cayley-Graves	$\mathbb{O}$	8	$\mathbb{O}_{C_8}$	16	C_8
Comp. Eisenstein	$\mathbb{O}$	8	$\mathbb{O}_{4A_2}$	24	$A_2 \oplus A_2 \oplus A_2 \oplus A_2$
Coupled Hurwitz	$\mathbb{O}$	8	$\mathbb{O}_{2D_4}$	48	$D_4 \oplus D_4$
Coxeter-Dickson	$\mathbb{O}$	8	$\mathbb{O}_{E_8}$	240	E_8

TABLE 1. Summary of all crystallographic sets of integer elements over division Hurwitz algebra $\mathbb{R}, \mathbb{C}, \mathbb{H}$ and $\mathbb{O}$. In the first column we indicated the name according to [Jo13]; then the related algebra in which the integral set is embedded; the dimension of the algebra; the notational symbol we introduced; the number of invertible unit elements; their algebraic structure as abelian group (real and complex case), non-abelian group (quaternionic case) and as Moufang loop (octonionic case); finally, in the last column the lattice associated with the integral set.

Although this definition is broad and widely accepted, it has two notable shortcomings. First, it can easily yield an infinite number of integral elements over Hurwitz division algebras, encompassing the real numbers $\mathbb{R}$ real numbers, the $\mathbb{C}$ complex numbers, the $\mathbb{H}$ quaternions and the $\mathbb{O}$ octonions. For example, for any natural number $m \in \mathbb{N}$, the subring $\mathbb{Z}\left[\sqrt{-m}\right]$ is a set of integral elements of $\mathbb{C}$. Second, its specificity -requiring the presence of the unit element- automatically excludes integral systems from non-unital algebras, such as para-octonions $p\mathbb{O}$ and Okubo algebra $\mathcal{O}$.

The first issue with Dickson's definition was addressed by Johnson [Jo13, Jo17, Jo18]. Inspired by the work of Coxeter [Co46], Johnson proposed that integral systems should inherently relate to lattices and, by extension, to crystallographic root systems. He then arrived to the following definition

Definition 1. A *basic system of integral elements* over a real algebra is given by a set of elements such that:

(1) The trace and the norm are integers. Specifically,
$$tr\,(x) = x + \overline{x} \in \mathbb{Z},$$
$$n\,(x) = x\overline{x} \in \mathbb{Z},$$
and where the conjugation is given by an involution of the algebra.
(2) The elements form a subring of the algebra, closed under multiplication, and comprising a set of invertible unit elements.
(3) The elements span a two-, four-, or eight-dimensional lattice embedded in $\mathbb{R}, \mathbb{C}, \mathbb{H}$ or $\mathbb{O}$.

Taking into account Johnson's refinements, the number of viable sets of integral elements dramatically diminishes, as detailed in Table 1).

1.1. **Complex integral sets.** In order to appreciate the algebraic richness of the octonionic case, it is instructive to begin with the simpler complex setting. The complex plane $\mathbb{C}$ admits

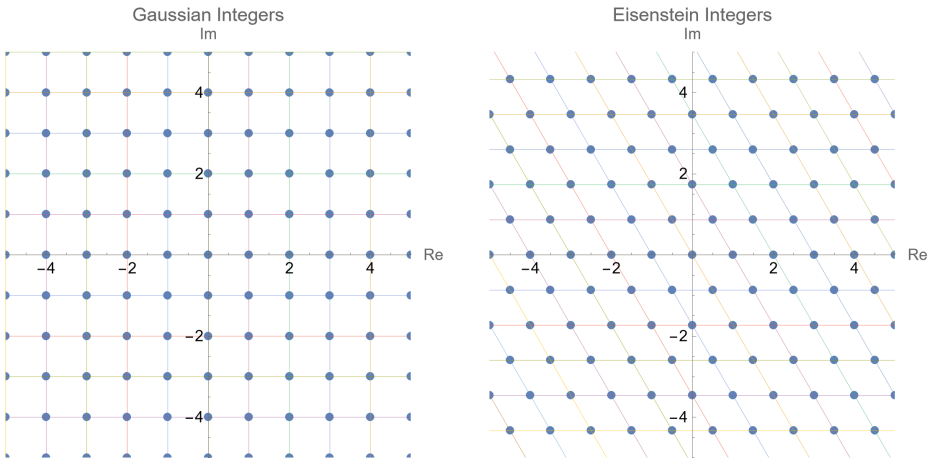

FIGURE 1.1. *On the left*: the Gaussian integers $\mathbb{Z}[i]$, forming a square lattice C_2 in the complex plane with four unit elements $\{\pm 1, \pm i\}$. *On the right*: the Eisenstein integers $\mathbb{Z}[\omega]$, forming a triangular lattice A_2 with six unit elements $\{\pm 1, \pm\omega, \pm\omega^2\}$.

two classical *crystallographic integral sets*, each corresponding to a different sublattice of the Gaussian plane.

The *Gaussian integers* $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$ form a ring under standard complex multiplication, with norm $n(a + bi) = a^2 + b^2 \in \mathbb{Z}$ and trace $tr(a + bi) = 2a \in \mathbb{Z}$. The four invertible unit elements are $\{+1, -1, +i, -i\}$, forming the cyclic group $\mathbb{Z}_4$. Their lattice is isometric to C_2, the square lattice.

The *Eisenstein integers* $\mathbb{Z}[\omega]$, where $\omega = e^{2\pi i/3} = \frac{-1+i\sqrt{3}}{2}$, constitute a second integral system. Here the norm is $n(a+b\omega) = a^2 - ab + b^2 \in \mathbb{Z}$ and the trace is $tr(a+b\omega) = 2a - b \in \mathbb{Z}$. The six unit elements are $\{\pm 1, \pm\omega, \pm\omega^2\}$, forming the cyclic group $\mathbb{Z}_6$, and their lattice is the triangular lattice A_2.

Remark 2. It is worth noting that $\sqrt{3}$ appears in the Eisenstein case through the imaginary part of ω. This is not merely a numerical coincidence: it reflects the underlying A_2 symmetry of the hexagonal lattice. The same coefficient appears in a useful rotation presentation of the Okubo construction. As we shall see, however, it belongs to that presentation and is not an intrinsic obstruction to integral Okubo orders over $\mathbb{Z}$.

1.2. **Quaternionic integral sets.** Moving to dimension four, the quaternions $\mathbb{H}$ admit three crystallographic integral sets, reflecting a richer algebraic structure.

The *Hamilton integers* $\mathbb{H}_{2C_2}$ consist of all quaternions $a_0 + a_1\mathrm{i} + a_2\mathrm{j} + a_3\mathrm{k}$ with $a_0, a_1, a_2, a_3 \in \mathbb{Z}$. Their eight unit elements $\{\pm 1, \pm\mathrm{i}, \pm\mathrm{j}, \pm\mathrm{k}\}$ form the quaternion group Q_8, and the lattice is $C_2 \oplus C_2$.

The *Hurwitz integers* $\mathbb{H}_{D_4}$ are the integral closure of $\mathbb{H}_{2C_2}$ in $\mathbb{H}$: they consist of all quaternions of the form $a_0 + a_1\mathrm{i} + a_2\mathrm{j} + a_3\mathrm{k}$ where either all $a_i \in \mathbb{Z}$ or all $a_i \in \mathbb{Z} + \frac{1}{2}$. They possess 24 unit elements forming the *binary tetrahedral group* $2T \cong \mathrm{SL}(2, 3)$, whose underlying lattice is D_4. The 24 unit elements correspond to the vertices of the 24-cell, the unique regular polytope in dimension four which is its own dual. In analogy to the octonionic case, the Hurwitz integers *enjoy* the property of being the maximal order in $\mathbb{H}$ over $\mathbb{Z}$.

The *Hybrid integers* $\mathbb{H}_{2A_2}$ are generated by combining two Eisenstein systems within $\mathbb{H}$ and possess 12 unit elements. Their lattice is $A_2 \oplus A_2$.

Remark 3. The Hurwitz integers $\mathbb{H}_{D_4}$ are, up to isomorphism, the unique maximal order in $\mathbb{H}$ over $\mathbb{Z}$. This maximality is the quaternionic analogue of the Coxeter-Dickson octonions being the maximal order in $\mathbb{O}$, a theme that recurs, with important modifications, in the Okubo setting where the product is changed while the coefficient ring remains $\mathbb{Z}$.

1.3. **Octonionic integral sets.** The octonionic case is considerably richer than its lower-dimensional counterparts. In analogy to the quaternionic case, there are multiple crystallographic integral sets in $\mathbb{O}$, nested by inclusion and distinguished by the number of unit elements they contain.

The *Cayley-Graves integers* $\mathbb{O}_{C_8}$ are the simplest octonionic integral set, consisting of octonions with all-integer coefficients in the standard basis $\{1, \mathrm{i}, \mathrm{j}, \mathrm{k}, \mathrm{l}, \mathrm{il}, \mathrm{jl}, \mathrm{kl}\}$. They possess 16 unit elements forming the Moufang loop $M_{16}(Q_8)$, and their lattice is C_8.

The *Compounded Eisenstein integers* $\mathbb{O}_{4A_2}$ are obtained by embedding four copies of the Eisenstein integers into $\mathbb{O}$ using compatible quaternionic subalgebras. They possess 24 unit elements forming the loop $M_{24}(4A_2)$, and their lattice is $A_2 \oplus A_2 \oplus A_2 \oplus A_2$.

The *Coupled Hurwitz integers* $\mathbb{O}_{2D_4}$ are obtained by coupling two copies of the Hurwitz quaternionic integers in a compatible manner. They possess 48 unit elements forming the loop $M_{48}(2D_4)$, and their lattice is $D_4 \oplus D_4$.

Finally, the maximal integral set is the *Coxeter-Dickson octonions* $\mathbb{O}_{E_8}$, to which we now turn. As we will see, this is the unique octonionic integral system whose lattice is the exceptional E_8, and it is the one against which the para-octonionic and Okubo products will be tested.

For our purposes, the integral set called Coxeter-Dickson octonions, and denoted here as $\mathbb{O}_{E_8}$, is of special importance. Following [Co46], we define as the octonionic elements $\mathbb{O}_{E_8}$, those of the form

$$x = a_0 + a_1\mathrm{i} + a_2\mathrm{j} + a_3\mathrm{k} + a_4\mathrm{h} + a_5\mathrm{ih} + a_6\mathrm{jh} + a_7\mathrm{kh}, \tag{1.3}$$

where $\mathrm{h} = (\mathrm{i} + \mathrm{j} + \mathrm{k} + \mathrm{l})/2$ and $a_0, ..., a_7 \in \mathbb{Z}$.

To see that this is indeed a set of integral elements under octonionic conjugation and multiplication we have first to show that the norm and the trace are integers. Indeed, a straightforward calculation yields to an explicit form for the norm

$$n(x) = a_0(a_0 - a_5 - a_6 - a_7) + a_1(a_1 + a_4 + a_6) + \tag{1.4}$$

$$+ a_2(a_2 + a_4 + a_5) + a_3(a_3 + a_4) + a_4^2 + a_5^2 + a_6^2 + a_7^2, \tag{1.5}$$

and for the trace

$$tr(x) = 2a_0 - a_5 - a_6 - a_7, \tag{1.6}$$

thus showing that $n(x), tr(x) \in \mathbb{Z}$. The unit invertible elements of the set are the 240 elements

$$\begin{array}{cc} \pm 1, \pm \mathrm{i}, \pm \mathrm{j}, \pm \mathrm{k}, & \pm \mathrm{l}, \pm \mathrm{il}, \pm \mathrm{jl}, \pm \mathrm{kl}, \\ \frac{1}{2}(\pm 1 \pm \mathrm{j} \pm \mathrm{k} \pm \mathrm{il}), & \frac{1}{2}(\pm \mathrm{i} \pm \mathrm{l} \pm \mathrm{jl} \pm \mathrm{kl}), \\ \frac{1}{2}(\pm 1 \pm \mathrm{k} \pm \mathrm{i} \pm \mathrm{jl}), & \frac{1}{2}(\pm \mathrm{j} \pm \mathrm{l} \pm \mathrm{kl} \pm \mathrm{il}), \\ \frac{1}{2}(\pm 1 \pm \mathrm{i} \pm \mathrm{j} \pm \mathrm{kl}), & \frac{1}{2}(\pm \mathrm{k} \pm \mathrm{l} \pm \mathrm{il} \pm \mathrm{jl}), \\ \frac{1}{2}(\pm 1 \pm \mathrm{il} \pm \mathrm{jl} \pm \mathrm{kl}), & \frac{1}{2}(\pm \mathrm{i} \pm \mathrm{j} \pm \mathrm{k} \pm \mathrm{l}), \\ \frac{1}{2}(\pm 1 \pm \mathrm{i} \pm \mathrm{l} \pm \mathrm{il}), & \frac{1}{2}(\pm \mathrm{j} \pm \mathrm{k} \pm \mathrm{jl} \pm \mathrm{kl}), \\ \frac{1}{2}(\pm 1 \pm \mathrm{j} \pm \mathrm{l} \pm \mathrm{jl}), & \frac{1}{2}(\pm \mathrm{k} \pm \mathrm{i} \pm \mathrm{kl} \pm \mathrm{il}), \\ \frac{1}{2}(\pm 1 \pm \mathrm{k} \pm \mathrm{l} \pm \mathrm{kl}), & \frac{1}{2}(\pm \mathrm{i} \pm \mathrm{j} \pm \mathrm{il} \pm \mathrm{jl}), \end{array} \tag{1.7}$$

$$\begin{pmatrix} e_0\mapsto 1 & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 \\ e_1 & -1 & e_4 & e_7 & -e_2 & e_6 & -e_5 & -e_3 \\ e_2 & -e_4 & -1 & e_5 & e_1 & -e_3 & e_7 & -e_6 \\ e_3 & -e_7 & -e_5 & -1 & e_6 & e_2 & -e_4 & e_1 \\ e_4 & e_2 & -e_1 & -e_6 & -1 & e_7 & e_3 & -e_5 \\ e_5 & -e_6 & e_3 & -e_2 & -e_7 & -1 & e_1 & e_4 \\ e_6 & e_5 & -e_7 & e_4 & -e_3 & -e_1 & -1 & e_2 \\ e_7 & e_3 & e_6 & -e_1 & e_5 & -e_4 & -e_2 & -1 \end{pmatrix}$$

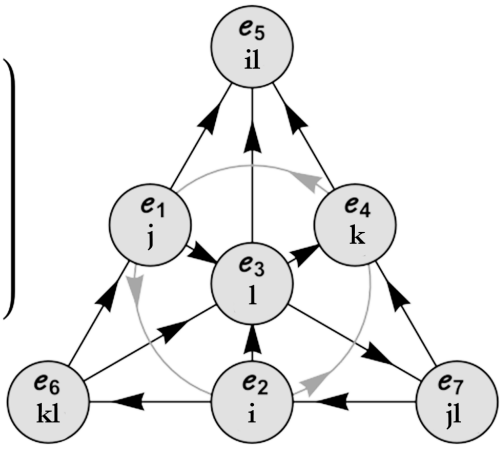

FIGURE 1.2. *On the left*: octonionic multiplication tables for the basis $\{e_0 = 1, e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$. *On the right*: a mnemonic representation on the Fano plane of the same octonionic multiplication rule with the equivalence with the Dickson notation $\{1, \mathrm{i}, \mathrm{j}, \mathrm{k}, \mathrm{l}, \mathrm{il}, \mathrm{jl}, \mathrm{kl}\}$ according to [Co46].

which identify the vertices of the Gosset polytope 4_{21}. Those are closed under octonionic multiplication (see Fig. 1.2). In fact, the elements in (1.7) form a well-known Moufang loop, here denoted as $M_{240}(E_8)$.

Finally, the isometry between the set of integral octonions $\mathbb{O}_{E_8}$ with the octonionic norm n and the E_8-lattice is a well-known result from a straightforward calculation and that can be explicitly found in [CoSm, Sec. 9] or [Koc89].

Remark 4. The shell structure of $\mathbb{O}_{E_8}$ is captured by the remarkable formula

$$|\mathbb{O}_{E_8}(n)| = 240 \sum_{d|n} d^3, \tag{1.8}$$

where $|\mathbb{O}_{E_8}(n)|$ denotes the number of elements of norm n in $\mathbb{O}_{E_8}$. In particular, the 240 unit elements ($n = 1$) are exactly the vertices of the Gosset polytope 4_{21}. This formula reflects the deep connection between E_8 and modular forms: the right-hand side is proportional to the Eisenstein series E_4.

2. Octonions, paraoctonions and the real Okubo algebra

Lattices are often constructed by taking discrete subsets of unit norm elements within an algebra and subsequently forming their integer span. Consequently, possessing a quadratic norm that respects the multiplicative structure of the algebra results in an intriguing interplay between the lattice elements and their algebraic properties. Normed algebras endowed with a norm that respect the multiplicative structure, i.e., $n(x \cdot y) = n(x)\, n(y)$, are called *composition algebras*.

Composition algebras are divided in *unital*, i.e. where exists a unit element 1 such that $x \cdot 1 = 1 \cdot x = x$, *para-unital*, i.e. where exists an involution $x \longrightarrow \overline{x}$ and an element called paraunit $\mathbf{1}$ such that $x \cdot \mathbf{1} = \mathbf{1} \cdot x = \overline{x}$, and, finally, *non-unital*, i.e. intended here for algebras that possess neither a unit element nor a paraunit element.

In fact, as a consequence of the Generalized Hurwitz Theorem [El18], over the real numbers, the finite-dimensional composition algebras fall into exactly sixteen isomorphism classes: seven

Hurwitz	**O.**	**C.**	**A.**	**Alt.**	**F.**		**p-Hurwitz**	**O.**	**C.**	**A.**	**Alt.**	**F.**
$\mathbb{R}$	Yes	Yes	Yes	Yes	Yes		$p\mathbb{R} \cong \mathbb{R}$	Yes	Yes	Yes	Yes	Yes
$\mathbb{C}$, $\mathbb{C}_s$	No	Yes	Yes	Yes	Yes		$p\mathbb{C}$, $p\mathbb{C}_s$	No	Yes	No	No	Yes
$\mathbb{H}$,$\mathbb{H}_s$	No	No	Yes	Yes	Yes		$p\mathbb{H}$,$p\mathbb{H}_s$	No	No	No	No	Yes
$\mathbb{O}$,$\mathbb{O}_s$	No	No	No	Yes	Yes		$p\mathbb{O}$,$p\mathbb{O}_s$	No	No	No	No	Yes

TABLE 2. *On the left,* we have summarized the algebraic properties, i.e. totally ordered (O), commutative (C), associative (A), alternative (Alt), flexible (F), of all Hurwitz algebras, namely $\mathbb{R}, \mathbb{C}, \mathbb{H}$ and $\mathbb{O}$ along with their split counterparts $\mathbb{C}_s, \mathbb{H}_s, \mathbb{O}_s$. *On the right*, we have summarized the algebraic properties of all para-Hurwitz algebras, namely $p\mathbb{R}, p\mathbb{C}, p\mathbb{H}$ and $p\mathbb{O}$ accompanied by their split counterparts $p\mathbb{C}_s, p\mathbb{H}_s, p\mathbb{O}_s$.

Property	$\mathbb{O}$	$p\mathbb{O}$	$\mathcal{O}$
Unital	Yes	No	No
Paraunital	Yes	Yes	No
Alternative	Yes	No	No
Flexible	Yes	Yes	Yes
Composition	Yes	Yes	Yes
Automorphism	G_2	G_2	SU (3)

TABLE 3. Synoptic table of the algebraic properties of octonions $\mathbb{O}$, paraoctonions $p\mathbb{O}$ and the real Okubo algebra $\mathcal{O}$.

are unital and called *Hurwitz algebras*, comprising the division algebras $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ along with their split companions $\mathbb{C}_s, \mathbb{H}_s, \mathbb{O}_s$; another seven are paraunital, closely related to the Hurwitz algebras and termed *para-Hurwitz algebras* (see Table 2); finally, there are two composition algebras, one division and one split, that are both non-unital and 8-dimensional, known as the *Okubo algebras* $\mathcal{O}$ and $\mathcal{O}_s$.

In this section we review the three division and composition algebras of dimension eight: the Hurwitz algebra of the octonions $\mathbb{O}$, the para-Hurwitz algebra of para-octonions $p\mathbb{O}$ and, finally, the Okubo algebra $\mathcal{O}$. The main objective of this section is summarized in Table 4 that synoptically illustrates the relationships between the product of the three algebras. The reason for limiting ourselves to the division algebras and not consider their split companions it will be clear in the next section, where we will require the closure under multiplication for the elements of unit norm.

An important final remark: these three algebras are deeply linked one another and it might be tempting to consider them equivalent in some sort of sense. Nevertheless, it is crucial to note that although is possible to switch from one algebra to another by altering the product definition, none of these algebras are isomorphic to the others. Specifically, the octonions $\mathbb{O}$ are alternative and unital, para-octonions $p\mathbb{O}$ are neither alternative nor unital but do have a para-unit, while the Okubo algebra $\mathcal{O}$ is non-alternative and only has idempotents elements. It is also worth highlighting that the Okubo algebra $\mathcal{O}$ is the least structured among these algebras (for a summary of the property of these algebras see Table 3).

2.1. **The algebra of octonions.** The algebra of octonions $\mathbb{O}$ is the only division Hurwitz algebra with a dimension of eight. We define the composition algebra of octonion $(\mathbb{O}, \cdot, n)$ as the eight dimensional real vector space with basis $\{e_0 = 1, e_1, ..., e_7\}$ with a bilinear product

encoded through the *Fano plane* and explained in Fig. 1.2. The resulting algebra is non-associative, non-commutative, but alternative and thus flexible. Given an element $x \in \mathbb{O}$ with decomposition

$$x = x_0 + \sum_{k=1}^{7} x_k \mathrm{e}_k, \tag{2.1}$$

the norm n is the obvious Euclidean one defined by

$$n(x) = x_0^2 + x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 + x_6^2 + x_7^2, \tag{2.2}$$

for which the conjugation results

$$\overline{x} = x_0 - \sum_{k=1}^{7} x_k i_k, \tag{2.3}$$

and therefore

$$n(x) = \overline{x} \cdot x, \tag{2.4}$$

as it happens for every Hurwitz algebra. Then a look at (2.2) shows that $n(x) = 0$ if and only if $x = 0$ and thus the inverse of a non-zero element of the octonions is easily found as

$$x^{-1} = \frac{\overline{x}}{n(x)}. \tag{2.5}$$

Also, from (2.4) we have that the octonionic inner product is given by

$$\langle x, y \rangle = x\overline{y} + y\overline{x}, \tag{2.6}$$

so that $\langle x, x \rangle = 2n(x)$.

2.2. **The algebra of para-octonions.** In unital composition algebras, as noted earlier, there exists a canonical involution, an order-two antihomomorphism known as *conjugation*. This can be defined using the orthogonal projection of the unit element as

$$x \mapsto \overline{x} = \langle x, 1 \rangle 1 - x. \tag{2.7}$$

This canonical involution has the distinctive property of being an antihomomorphism with respect to the product, i.e., $\overline{x \cdot y} = \overline{y} \cdot \overline{x}$, and the basic property with the norm of $x \cdot \overline{x} = n(x) 1$.

Given the order-two antihomomorphism of the conjugation over the Hurwitz algebra of octonions $(\mathbb{O}, \cdot, n)$ we can now obtain a para-Hurwitz algebra defining a new product

$$x \bullet y = \overline{x} \cdot \overline{y}, \tag{2.8}$$

for every $x, y \in \mathbb{O}$. The new algebra $(\mathbb{O}, \bullet, n)$ is again a composition algebra, in fact a para-Hurwitz algebra, called para-octonions and denoted with $p\mathbb{O}$. As expected the algebra does not have a unit but only a para-unit, i.e. $1 \in p\mathbb{O}$ such that $1 \bullet x = \overline{x}$. Finally, it is worth noting that also the algebra of para-octonions is a division algebra since if

$$x \bullet y = \overline{x} \cdot \overline{y} = 0, \tag{2.9}$$

then either $\overline{x}$ or $\overline{y}$ are zero and thus implying that either x or y are zero.

2.3. **The Okubo algebra.** The Petersson construction depends not only on the real automorphism, but also on its position with respect to a chosen integral order. This distinction is invisible over $\mathbb{R}$ and becomes essential over $\mathbb{Z}$. We begin with the rotation presentation used in the our previous works such as [CMZ24, CMZ25, MCZ25]. To avoid confusion, we denote it by ρ:

$$\begin{array}{ll} \rho\left(\mathrm{e}_k\right) & =\mathrm{e}_k, k=0,1,3,7, \\ \rho\left(\mathrm{e}_2\right) & =-\frac{1}{2}\left(\mathrm{e}_2-\sqrt{3}\mathrm{e}_5\right), \\ \rho\left(\mathrm{e}_5\right) & =-\frac{1}{2}\left(\mathrm{e}_5+\sqrt{3}\mathrm{e}_2\right), \\ \rho\left(\mathrm{e}_4\right) & =-\frac{1}{2}\left(\mathrm{e}_4-\sqrt{3}\mathrm{e}_6\right), \\ \rho\left(\mathrm{e}_6\right) & =-\frac{1}{2}\left(\mathrm{e}_6+\sqrt{3}\mathrm{e}_4\right). \end{array} \tag{2.10}$$

This is an order-three automorphism of $\mathbb{O}$. It fixes the quaternion subalgebra generated by $\{1, \mathrm{e}_1, \mathrm{e}_3, \mathrm{e}_7\}$. This subalgebra contains $w_\rho = (-1+\sqrt{3}\mathrm{e}_1)/2$, with $w_\rho^2 + w_\rho + 1 = 0$. Therefore [El18a, Theorem 5.1] shows that its Petersson isotope is a real Okubo algebra. There is no algebraic error in this construction. The problem is arithmetic: already $\rho(\mathrm{e}_2)$ does not belong to the Cayley–Graves order. Moreover, an explicit product of Coxeter–Dickson basis elements will show that its Petersson product does not preserve $\mathbb{O}_{E_8}$ either.

The two real planes generated by $\{\mathrm{e}_2, \mathrm{e}_5\}$ and $\{\mathrm{e}_4, \mathrm{e}_6\}$ explain the displayed coefficients. On both planes ρ acts as a rotation by $2\pi/3$, with the same matrix as multiplication by the primitive cubic root $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$.

For integral questions we use instead the following signed permutation of the Fano basis:

$$\begin{array}{lll} \tau(1)=1, & \tau(\mathrm{e}_3)=\mathrm{e}_3, & \\ \tau(\mathrm{e}_1)=\mathrm{e}_2, & \tau(\mathrm{e}_2)=-\mathrm{e}_4, & \tau(\mathrm{e}_4)=-\mathrm{e}_1, \\ \tau(\mathrm{e}_5)=-\mathrm{e}_7, & \tau(\mathrm{e}_7)=\mathrm{e}_6, & \tau(\mathrm{e}_6)=-\mathrm{e}_5. \end{array} \tag{2.11}$$

Direct inspection of the seven oriented lines of the Fano plane gives $\tau(x \cdot y) = \tau(x) \cdot \tau(y)$, while the two signed three-cycles give $\tau^3 = \mathrm{id}$. Thus $\tau \in \mathrm{G}_2$ is an order-three automorphism defined over $\mathbb{Z}$.

It remains to determine which symmetric composition algebra is obtained. Set

$$a = -\mathrm{e}_1 - \mathrm{e}_2 + \mathrm{e}_4, \qquad b = -\mathrm{e}_5 + \mathrm{e}_6 + \mathrm{e}_7. \tag{2.12}$$

The fixed subalgebra is

$$\mathrm{Fix}(\tau) = \mathrm{span}_{\mathbb{R}}\{1, \mathrm{e}_3, a, b\}. \tag{2.13}$$

Indeed, the Fano multiplication gives

$$a^2 = b^2 = -3, \qquad ab = 3\mathrm{e}_3 = -ba, \tag{2.14}$$

so this fixed subalgebra is isomorphic to the quaternions. In addition, $w = (-1+a)/2$ satisfies $w^2 + w + 1 = 0$. By the classification of Petersson algebras associated with order-three automorphisms [El18a, Theorem 5.1], the corresponding isotope is an Okubo algebra, and not a para-octonionic algebra. Since the norm is positive definite, it is the compact real division Okubo algebra.

We can now define its Petersson product by

$$x * y = \tau\left(\overline{x}\right) \cdot \tau^2\left(\overline{y}\right), \tag{2.15}$$

for every $x, y \in \mathbb{O}$. Note that the element 1 is an idempotent, since $1 * 1 = 1$, but it is not a unit or a paraunit. It is worth noting that this Okubo algebra is a division algebra: if

$$x * y = \tau\left(\overline{x}\right) \cdot \tau^2\left(\overline{y}\right) = 0, \tag{2.16}$$

then either $\tau\left(\overline{x}\right)$ or $\tau^2\left(\overline{y}\right)$ are zero and since τ is a homomorphism, either x or y are zero.

Unlike the case of the para-octonions it is worth presenting here an independent way of realising the Okubo algebra. In fact, the following realisation was the one independently found by Okubo ten years after Petersson. Following [Ok78] and [EM90], we define the real Okubo Algebra $\mathcal{O}$ as the set of three by three Hermitian traceless matrices over the complex numbers $\mathbb{C}$ with the following bilinear product

$$x * y = \mu \cdot xy + \overline{\mu} \cdot yx - \frac{1}{3}\mathrm{Tr}\,(xy)\, I_3, \tag{2.17}$$

where $\mu = 1/6\left(3 + \mathrm{i}\sqrt{3}\right)$ and the juxtaposition is the ordinary associative product between matrices. It is worth noting that (2.17) can be seen as a modification of the Jordanian product. Indeed, setting $\mu = 1/2$ and neglecting the last term, we retrieve the usual Jordan product over Hermitian traceless matrices, i.e.

$$x \circ y = \frac{1}{2}xy + \frac{1}{2}yx. \tag{2.18}$$

Nevertheless, Hermitian traceless matrices are not closed under such product, thus requiring the additional term $-1/3\mathrm{Tr}\,(xy)$ for the closure of the algebra. Indeed, setting in (2.17) $\mathrm{Im}\mu = 0$, one retrieves from the traceless part of the exceptional Jordan algebra $\mathfrak{J}_3\,(\mathbb{C})$, whose derivation Lie algebra is $\mathfrak{su}\,(3)$.

Analyzing (2.17), it becomes evident that the resulting algebra is neither unital, associative, nor alternative. Nonetheless, $\mathcal{O}$ is a *flexible* algebra, i.e.

$$x * (y * x) = (x * y) * x, \tag{2.19}$$

which will turn out to be an even more useful property than alternativity in the definition of the projective plane. Even though the Okubo algebra is not unital, it does have idempotents, i.e. $e * e = e$, such as

$$e = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \tag{2.20}$$

that together with

$$\begin{aligned} \mathrm{e}_1 &= \sqrt{3}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \mathrm{e}_2 &= \sqrt{3}\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \\ \mathrm{e}_3 &= \sqrt{3}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, & \mathrm{e}_4 &= \sqrt{3}\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ \mathrm{e}_5 &= \sqrt{3}\begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \mathrm{e}_6 &= \sqrt{3}\begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \\ \mathrm{e}_7 &= \sqrt{3}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \end{aligned} \tag{2.21}$$

form a basis for $\mathcal{O}$ that has real dimension 8. It is worth noting that the choice of the idempotent e as in (2.20) does not yield to any loss of generality for the subsequent development of our work since all idempotents form a single orbit under the full automorphism group SU(3) (see [El15, Thm. 20]); the cyclic subgroup $\langle\tau\rangle \cong \mathbb{Z}_3$ alone does not suffice to connect all idempotents. The choice of this special basis is useful for the matrix realization, but its orthogonality with respect to the norm in (2.22) is a separate Gram-matrix check and will not be used as an arithmetic input below. Through a special bijective map between Okubo algebra and octonions the elements of the basis $\{e, \mathrm{e}_1, ..., \mathrm{e}_7\}$ correspond to the octonionic ones previously defined.

In this realisation a direct way of defining the quadratic form n over Okubo algebra, is through the following

$$n(x) = \frac{1}{6}\mathrm{Tr}\left(x^2\right), \tag{2.22}$$

for every $x \in \mathcal{O}$. It is straightforward to see that the *norm* has signature $(8,0)$ and is a composition norm over the Okubo algebra; the associated symmetric bilinear form $\langle x, y\rangle$ is associative in the sense $\langle x \cdot z, y\rangle = \langle x, z \cdot y\rangle$.

2.4. **Okubo algebra, octonions and para-octonions.** An important feature of the Okubo algebra $\mathcal{O}$ is its interplay with the algebra of octonions $\mathbb{O}$. Indeed, octonions and the Okubo algebra are linked one another in such a way that we can easily pass from one to the other simply changing the definition of the bilinear product over the vector space of the algebra. Let us consider the Kaplansky's trick we introduced earlier and let us define a new product over the Okubo algebra $\mathcal{O}$ as

$$x \cdot y = (e * x) * (y * e), \tag{2.23}$$

where $x, y \in \mathcal{O}$ and e is an idempotent of $\mathcal{O}$. Given that $e * e = e$ and $n(e) = 1$, the element e acts as a left and right identity, i.e.

$$x \cdot e = e * x * e = n(e)\, x = x, \tag{2.24}$$

$$e \cdot x = e * x * e = n(e)\, x = x. \tag{2.25}$$

Moreover, since Okubo algebra is a composition algebra, the same norm n enjoys the following relation

$$n(x \cdot y) = n((e * x) * (y * e)) = n(x)\, n(y), \tag{2.26}$$

which means that $(\mathcal{O}, \cdot, n)$ is a unital composition algebra of real dimension 8. Since it is also a division algebra, then it must be isomorphic to that of octonions $\mathbb{O}$ as noted by Okubo himself [Ok78, Ok78c]. On the other hand, al already noticed, if we consider the order three automorphism of the octonions in (2.11), the Okubo algebra is then realised as a Petersson algebra from the octonions setting

$$x * y = \tau(\overline{x}) \cdot \tau^2(\overline{y}). \tag{2.27}$$

Note that (2.11) is formulated assuming the knowledge of the octonionic product. Reading the same maps as Okubic maps we then have the notable relation, i.e.

$$\overline{x} = ((x * e) * e) * e, \tag{2.28}$$

$$\tau(x) = (((x * e) * e) * e) * e, \tag{2.29}$$

so that, in fact, the two maps are linked one another, i.e.

$$\tau(x) = \overline{x} * e \tag{2.30}$$

$$\overline{x} = \tau(e * x). \tag{2.31}$$

While these maps are intertwined, it is important to highlight their distinct impacts on the algebra's structure. While τ is an automorphism for both Okubo algebra $\mathcal{O}$ and octonions $\mathbb{O}$, $\overline{x}$ do not respect the algebraic structure of the Okubo algebra $\mathcal{O}$, since it is not an automorphism nor an anti-automorphism with respect to the Okubo product, while it is an anti-homomorphism over octonions $\mathbb{O}$.

Algebra	$(\mathcal{O}, *, n)$	$(p\mathbb{O}, \bullet, n)$	$(\mathbb{O}, \cdot, n)$
$x * y$	$x * y$	$\tau(x) \bullet \tau^2(y)$	$\tau(\overline{x}) \cdot \tau^2(\overline{y})$
$x \bullet y$	$\tau^2(x) * \tau(y)$	$x \bullet y$	$\overline{x} \cdot \overline{y}$
$x \cdot y$	$(e * x) * (y * e)$	$(\mathbf{1} \bullet x) \bullet (y \bullet \mathbf{1})$	$x \cdot y$

TABLE 4. In this table we see how to obtain the Okubo product $*$, the para-octonionic product $\bullet$ and the octonionic product $\cdot$ from Okubo algebra $(\mathcal{O}, *, n)$, para-octonions $(p\mathbb{O}, \bullet, n)$ and octonions $(\mathbb{O}, \cdot, n)$ respectively

The scenario with para-octonions, $p\mathbb{O}$ is more straightforward. By definition, para-octonions are obtainable from octonions $\mathbb{O}$ through

$$x \bullet y = \overline{x} \cdot \overline{y}, \tag{2.32}$$

while, on the other hand, octonions $\mathbb{O}$ are obtainable from para-octonions $p\mathbb{O}$ through the aid of the para-unit $1 \in p\mathbb{O}$, such that

$$x \cdot y = (1 \bullet x) \bullet (y \bullet 1) \tag{2.33}$$

$$= \overline{x} \bullet \overline{y} = x \cdot y. \tag{2.34}$$

The new algebra $(p\mathbb{O}, \cdot, n)$ is again an eight-dimensional composition algebra which is also unital and division and thus, for Hurwitz theorem, isomorphic to that of octonions $\mathbb{O}$. Moreover, since $\tau(\overline{x}) = \overline{\tau(x)}$, we also have that the Okubic algebra is obtainable from the para-Hurwitz algebra with the introduction of a Petersson-like product, i.e.

$$x * y = \tau(x) \bullet \tau^2(y). \tag{2.35}$$

We thus have that all algebras are obtainable one from the other as summarized in Table 4. Nonetheless, it is vital to note that while transitioning from one algebra to another is feasible, these algebras are not isomorphic. For example, while the octonions $\mathbb{O}$ are alternative and unital, para-octonions $p\mathbb{O}$ are neither alternative nor unital but do have a para-unit. In contrast, the Okubo algebra $\mathcal{O}$ is non-alternative and only contains idempotent elements.

3. Integral systems for the three products

The unit in Johnson's definition must be treated with care when the product is changed. For the para-octonions the element 1 is a paraunit, while for the Petersson model of the Okubo algebra it is an idempotent. The norm and the Euclidean lattice, on the other hand, do not change. This suggests the following simple adaptation.

Definition 5. Let A be an eight-dimensional real composition algebra with product $\circ$, norm n, and a chosen nonzero idempotent e. A $\mathbb{Z}$-*integral system relative to* e is a free $\mathbb{Z}$-module $L \subset A$ of rank eight such that:

(1) $e \in L$ and L is closed under $\circ$;
(2) $n(x)$ and $\mathrm{tr}_e(x) = \langle x, e \rangle$ belong to $\mathbb{Z}$ for every $x \in L$;
(3) L is a discrete Euclidean lattice for the bilinear form associated with n.

The following observation reduces the closure problem to the arithmetic action of the automorphism.

Proposition 6 (Closure criterion)**.** *Let L be an octonionic $\mathbb{Z}$-order which is stable under conjugation, and let $\sigma \in \mathrm{Aut}(\mathbb{O})$ satisfy $\sigma(L) = L$. Then L is closed under both*

$$x \bullet y = \overline{x} \cdot \overline{y}, \qquad x *_\sigma y = \sigma(\overline{x}) \cdot \sigma^2(\overline{y}). \tag{3.1}$$

If $1 \in L$, the norm and relative trace conditions are the same as for the original octonionic order.

Proof. For $x, y \in L$, conjugation gives $\overline{x}, \overline{y} \in L$. The assumption on σ then gives $\sigma(\overline{x}), \sigma^2(\overline{y}) \in L$, and octonionic closure proves both assertions. All three products have the same composition norm. Moreover, relative to the idempotent 1, the trace is $\mathrm{tr}_1(x) = \langle x, 1\rangle$, which is the usual integral trace functional on the orders considered below. □

3.1. **Why the old rotation does not work integrally.** Let $*_\rho$ denote the Petersson product associated with (2.10). Since $\rho(\mathrm{e}_2)$ has nonintegral coordinates, the Cayley–Graves order is not ρ-stable. The same problem occurs for the Coxeter–Dickson order, although it is less immediate.

Write $\mathrm{h} = (\mathrm{e}_1 + \mathrm{e}_2 + \mathrm{e}_3 + \mathrm{e}_4)/2$ and use the Coxeter–Dickson basis

$$\begin{aligned} &b_0 = 1, \quad b_1 = \mathrm{e}_1, \quad b_2 = \mathrm{e}_2, \quad b_3 = \mathrm{e}_3, \\ &b_4 = \mathrm{h}, \quad b_5 = \mathrm{e}_1\mathrm{h}, \quad b_6 = \mathrm{e}_2\mathrm{h}, \quad b_7 = \mathrm{e}_3\mathrm{h}. \end{aligned} \tag{3.2}$$

An exact calculation gives

$$\begin{aligned} b_0 *_\rho b_2 = &-\frac{3}{2}\sqrt{3}\, b_0 + \frac{1}{2}\sqrt{3}\, b_1 + \left(\frac{1}{2} - \frac{1}{2}\sqrt{3}\right) b_2 \\ &- \sqrt{3}\, b_5 - \sqrt{3}\, b_6 - \sqrt{3}\, b_7. \end{aligned} \tag{3.3}$$

Thus $L_E *_\rho L_E \not\subseteq L_E$. This is why the old automorphism does not work for the integral problem.

Remark 7. The conclusion is deliberately limited to the chosen presentation. The automorphism ρ is valid and $*_\rho$ gives rise to the compact real Okubo algebra. Extending coefficients to $\mathbb{Z}[\sqrt{3}]$ or rescaling a basis may produce orders adapted to ρ, but this does not prove that $\sqrt{3}$ is intrinsic to Okubo integrality. The new automorphism (2.11) removes this presentation-dependent obstruction.

3.2. **Four orders preserved by the integral automorphism.** We now choose concrete representatives of the four octonionic integral systems introduced in Sec. 1. The Cayley–Graves order is

$$L_C = \mathbb{Z}\{1, \mathrm{e}_1, \ldots, \mathrm{e}_7\}. \tag{3.4}$$

A compatible compounded Eisenstein order L_A has basis

$$\begin{aligned} &p_0 = 1, && p_1 = \frac{-1 + \mathrm{e}_5 - \mathrm{e}_6 - \mathrm{e}_7}{2}, \\ &p_2 = \frac{-\mathrm{e}_2 + \mathrm{e}_4 + \mathrm{e}_5 + \mathrm{e}_6}{2}, && p_3 = \frac{-\mathrm{e}_1 + \mathrm{e}_2 - \mathrm{e}_3 - \mathrm{e}_4}{2}, \\ &p_4 = \frac{\mathrm{e}_2 + \mathrm{e}_3 + \mathrm{e}_5 + \mathrm{e}_7}{2}, && p_5 = \frac{\mathrm{e}_1 - \mathrm{e}_2 - \mathrm{e}_3 - \mathrm{e}_4}{2}, \\ &p_6 = \frac{\mathrm{e}_1 + \mathrm{e}_2 + \mathrm{e}_6 - \mathrm{e}_7}{2}, && p_7 = \frac{-\mathrm{e}_1 - \mathrm{e}_2 + \mathrm{e}_3 - \mathrm{e}_4}{2}. \end{aligned} \tag{3.5}$$

A compatible coupled Hurwitz order L_D has basis

$$\begin{aligned} &d_0 = 1, \quad d_1 = \mathrm{e}_1, \quad d_2 = \mathrm{e}_2, \quad d_3 = \frac{1 + \mathrm{e}_1 + \mathrm{e}_2 + \mathrm{e}_4}{2}, \\ &d_4 = \mathrm{e}_3, \quad d_5 = \mathrm{e}_1\mathrm{e}_3, \quad d_6 = \mathrm{e}_2\mathrm{e}_3, \quad d_7 = d_3\mathrm{e}_3. \end{aligned} \tag{3.6}$$

Finally, $L_E = \mathbb{O}_{E_8}$ is the Coxeter–Dickson order with basis (3.2). Their metric lattices are respectively C_8, A_2^4, D_4^2, and E_8.

Order	Metric lattice	Gram determinant	$\lvert L(1)\rvert$	Okubo idempotents
L_C	C_8	2^8	16	4
L_A	A_2^4	3^4	24	3
L_D	D_4^2	4^2	48	9
L_E	E_8	1	240	12

TABLE 5. The four lattices preserved by the integral Okubo product. The Gram form is $\langle x, x\rangle = 2n(x)$.

Proposition 8 (Preservation of the four integral systems)**.** *The four orders L_C, L_A, L_D, and L_E are stable under the automorphism τ in (2.11). Consequently, all four are closed under the para-octonionic product and under the Okubo product*

$$x * y = \tau(\overline{x}) \cdot \tau^2(\overline{y}). \tag{3.7}$$

They give $\mathbb{Z}$-integral systems for the three composition algebras.

Proof. The displayed bases are closed under octonionic multiplication and conjugation; this is the usual order property. For L_C, stability under τ follows immediately because (2.11) is a signed permutation. For the other three orders it can be checked without changing coefficient rings. On the A_2^4 basis one obtains

$$\begin{array}{llll} \tau(p_0) = p_0, & \tau(p_1) = p_1, & \tau(p_2) = -p_4 - p_5, & \tau(p_3) = p_5, \\ \tau(p_4) = p_6 + p_7, & \tau(p_5) = -p_7, & \tau(p_6) = -p_2, & \tau(p_7) = -p_3. \end{array} \tag{3.8}$$

On the D_4^2 basis the action is

$$\begin{array}{lll} \tau(d_0) = d_0, \quad \tau(d_1) = d_2, & & \tau(d_2) = d_0 + d_1 + d_2 - 2d_3, \\ \tau(d_3) = d_0 + d_2 - d_3, \quad \tau(d_4) = d_4, & & \tau(d_5) = d_6, \\ \tau(d_6) = d_4 + d_5 + d_6 - 2d_7, & & \tau(d_7) = d_4 + d_6 - d_7. \end{array} \tag{3.9}$$

Finally, on the Coxeter–Dickson basis one has

$$\begin{array}{lll} \tau(b_0) = b_0, & \tau(b_1) = b_2, & \tau(b_2) = b_1 + b_2 + b_3 - 2b_4, \\ \tau(b_3) = b_3, & \tau(b_4) = b_2 + b_3 - b_4, & \tau(b_5) = -2b_1 - b_2 - b_3 + 2b_4 + b_6, \\ \tau(b_6) = b_0 + b_2 + b_5 + b_6 + b_7, & & \\ \tau(b_7) = -2b_0 + 2b_1 + b_2 + b_3 & & \\ \qquad - 2b_4 - 2b_6 - b_7. & & \end{array} \tag{3.10}$$

All coefficients are integers. Since $\tau^3 = \mathrm{id}$, these inclusions are equalities. Proposition 6 now proves closure for both products, while the norm and trace conditions are inherited from the original octonionic orders. □

The metric information and a first multiplicative invariant are summarized in Table 5. Here $L(1)$ denotes the set of elements of norm one, and the last column counts the elements satisfying $x * x = x$. The counts follow by solving the positive definite equation $n(x) = 1$ in each displayed basis and testing the resulting finite set under the exact product (3.7).

4. The octonionic, para-octonionic and Okubo E_8 structures

Proposition 8 reveals a remarkable phenomenon on E_8: one additive lattice L_E carries three different products.

Proposition 9 (One lattice, three arithmetic algebras)**.** *The identity map on L_E is an isometry among the octonionic, para-octonionic, and Okubo integral systems. In particular, all three systems have the same Gram matrix, the same* 240 *roots, and the same Gosset polytope* 4_{21}*. They are pairwise non-isomorphic as algebras.*

Proof. The para-Hurwitz and Petersson constructions change only the multiplication. Their norm is the original octonionic norm, so the additive group and the Gram form are exactly the same in all three cases. Hence the vectors of norm one are the same 240 vectors listed in (1.7). On the other hand, the octonionic product is unital and alternative; the para-octonionic product is non-unital and non-alternative, but has the paraunit 1; and the Okubo product is neither unital nor para-unital and is non-alternative. These properties are invariant under algebra isomorphisms. □

The difference can already be seen on the common root set. With octonionic multiplication the 240 roots form the Moufang loop $M_{240}(E_8)$ and 1 is its identity. With the para-octonionic multiplication the same set is the para-isotope of this Moufang loop: it has the paraunit 1, satisfying $1 \bullet x = x \bullet 1 = \overline{x}$, but it has no identity. With the Okubo multiplication the same finite set is again closed, but has neither an identity nor a paraunit. In both non-unital cases the root set is a quasigroup. Indeed, for a fixed root u, left and right multiplication by u are injective because the corresponding real composition algebra is a division algebra; on a finite set they are therefore bijective.

The idempotents give a simple numerical distinction. The octonionic division algebra has only the nonzero idempotent 1. Among the common 240 roots, 57 are para-octonionic idempotents: besides 1, they are the 56 integral roots of the form

$$x = -\frac{1}{2} + v, \qquad v \in \mathrm{Im}(\mathbb{O}), \qquad n(v) = \frac{3}{4}. \tag{4.1}$$

For the Okubo product there are 12 integral idempotents. Thus the geometry is unchanged, but the three multiplications organise its vertices in different ways.

There is also a difference between continuous and arithmetic symmetries. The real octonion and para-octonion algebras both have automorphism group G_2, whereas the real Okubo algebra has automorphism group $\mathrm{SU}(3)$. These full compact groups do not preserve a chosen E_8 order in general. For the present embeddings the relevant finite groups are

$$G_E^{\mathrm{oct}} = G_E^{\mathrm{poct}} = \{g \in \mathrm{G}_2 : g(L_E) = L_E\}, \tag{4.2}$$

$$G_E^{\mathrm{Ok}} = \{g \in \mathrm{SU}(3) : g(L_E) = L_E\}. \tag{4.3}$$

The first equality follows because octonion automorphisms commute with conjugation and therefore preserve the para-product. Thus the octonionic and para-octonionic arithmetic stabilizers coincide.

Every octonion automorphism preserving L_E and commuting with τ preserves the Petersson product. Consequently,

$$C_{G_E^{\mathrm{oct}}}(\tau) \subseteq G_E^{\mathrm{Ok}}, \tag{4.4}$$

and τ itself gives a nontrivial element of the Okubo arithmetic stabilizer. Determining whether this inclusion accounts for the complete stabilizer is a separate finite group problem.

Remark 10. The rotation ρ and the integral automorphism τ both give real Okubo algebras, but they define different arithmetic embeddings with respect to the standard octonionic orders. For ρ one sees $\sqrt{3}$ and loses the four lattices; for τ the matrices are integral and all four lattices are preserved. The algebraic isomorphism class alone therefore does not determine the integral order.

5. Conclusions and Future Developments

In this work we have shown that the obstruction found in the old Okubo presentation was not intrinsic. The rotation automorphism ρ is algebraically correct and produces the compact real Okubo algebra, but it does not preserve the classical integral octonion orders. The signed automorphism τ is again of order three, has a quaternionic fixed subalgebra, and its Petersson isotope is therefore an Okubo algebra. Unlike ρ, it is defined over $\mathbb{Z}$ and preserves the four systems C_8, A_2^4, D_4^2, and E_8.

The most important consequence is that the Coxeter–Dickson lattice is itself an integral order for all three products. The octonionic, para-octonionic, and Okubo E_8 structures have the same Euclidean lattice, the same 240 roots, and the same Gosset polytope. They differ only after multiplication is introduced: the octonionic roots form a unital Moufang loop, the para-octonionic roots form a quasigroup with a paraunit and 57 idempotents, and the Okubo roots form a quasigroup with neither unit nor paraunit and 12 idempotents.

The main similarities and differences are summarized in Table 6.

Property	Octonions	Para-octonions	Okubo algebra
Underlying lattice	E_8	E_8	E_8
Product	$x \cdot y$	$\overline{x} \cdot \overline{y}$	$\tau(\overline{x}) \cdot \tau^2(\overline{y})$
Unit structure	Unit 1	Paraunit 1	Neither unit nor paraunit
Algebraic laws	Alternative and flexible	Non-alternative and flexible	Non-alternative and flexible
Real automorphism group	G_2	G_2	SU(3)
Norm-one roots	Moufang loop $M_{240}(E_8)$	Para-isotope of $M_{240}(E_8)$	Okubo quasigroup
Integral idempotents in E_8	1	57	12

Table 6. Comparison of the three integral algebra structures carried by the same E_8 lattice.

It would be interesting to determine the full group G_E^{Ok}, its orbits on the 240 roots and on the 12 idempotents, and its relation with the centralizer of τ in the octonionic arithmetic group. A systematic study of other order-three automorphisms preserving maximal octonion orders may also clarify how many non-isomorphic integral Okubo products can live on the same E_8 lattice.

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Grupo de Física Matemática, Instituto Superior Técnico, Av. Rovisco Pais, 1049-001 Lisboa, Portugal

Email address: danielecorradetti@tecnico.ulisboa.pt